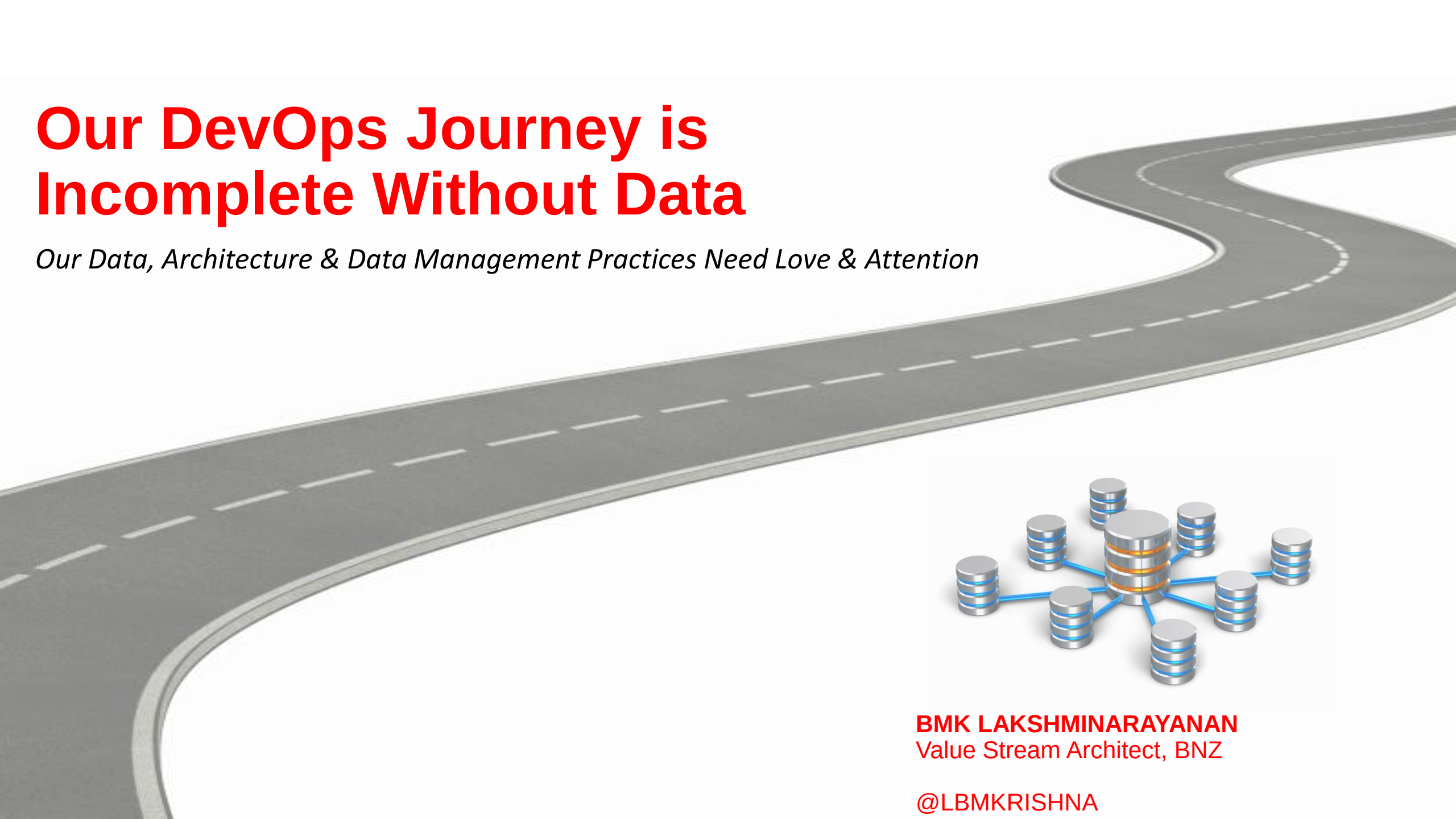
Our DevOps Journey is Incomplete Without Data

Our Data, Architecture & Data Management Practices Need Love & Attention



BMK LAKSHMINARAYANAN
Value Stream Architect, BNZ

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Data is the lifeblood of the system of the business

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Flow

Problem Context	Context, Statistics, Impact
Challenges	Ownership, Silo, Locking, Standards
Solution Approach	Education, Conventions, Architecture, Refactor
Recommendations	Useful Resources
Remaining Challenges	Call for Action

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ABOUT ME

BMK Lakshminarayanan

Solutions Architect | DevOps Advocate

MISSION

Make your life better,
Make others life better,
Make the world better

Community

DEVOPS DAYS

CLOUD NATIVE SUMMIT

CLOUD NATIVE COMPUTING FOUNDATION AMBASSADOR

AMBASSADOR DevOps INSTITUTE Human of DevOps

CD.FOUNDATION

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Understanding Our Problem

Problem Context

Continuous Delivery, DevOps & Data

Problem Context

- Manual database deployments continue to be a Software delivery bottleneck
- Speeding up database releases are most difficult
- Neglecting the data in our DevOps, Continuous Delivery processes has significant “Business Impact”
- Data issues continue to slow us down, trouble our efforts to continuously integrate, test & deploy

Statistics

46% Extremely or very difficult to speed database releases

23% High risk of deployment failure & extra downtime due to database changes

20% We are slow due to traditional siloed database development practices

57% Application changes require corresponding database changes

Source :https://www.devopsdigest.com/the-state-of-database-deployments-in-application-delivery-2019

Yesterday's DBA

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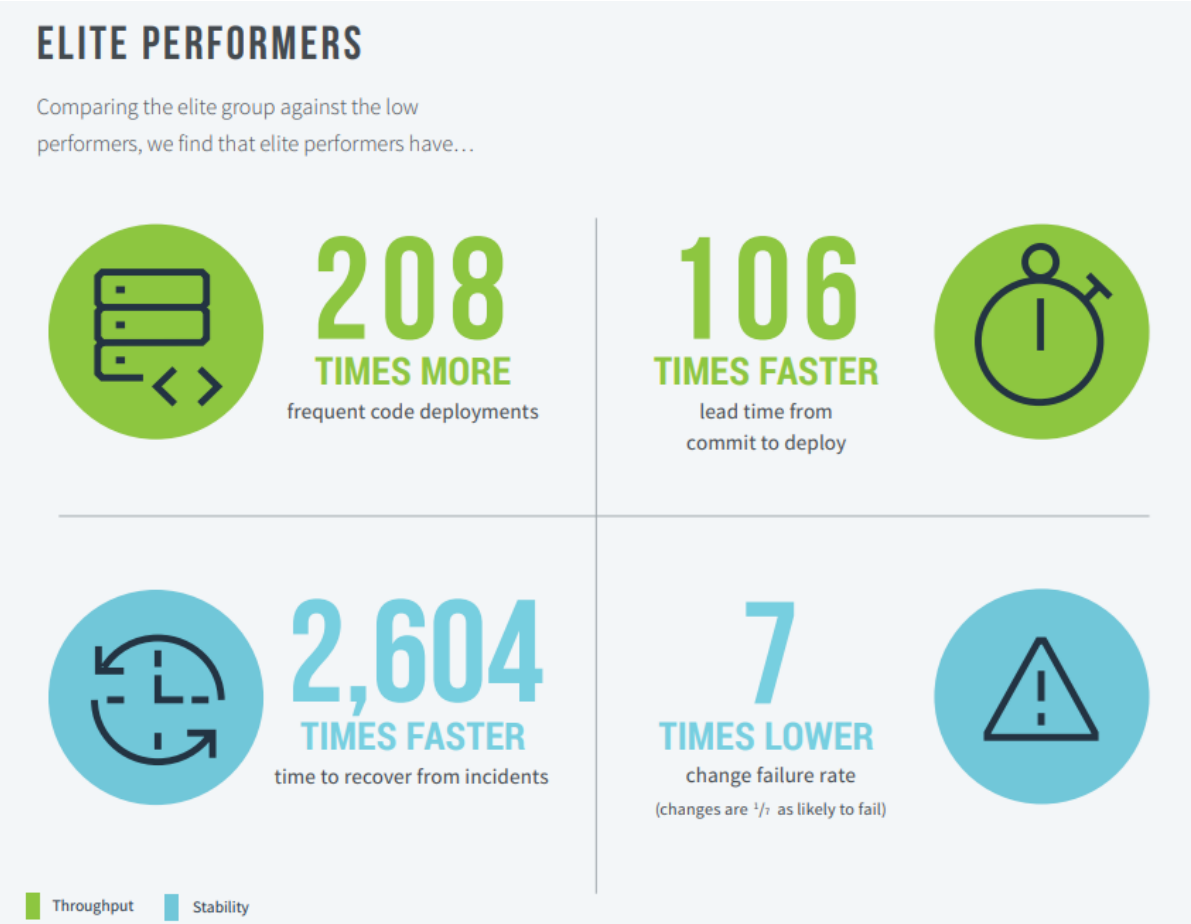
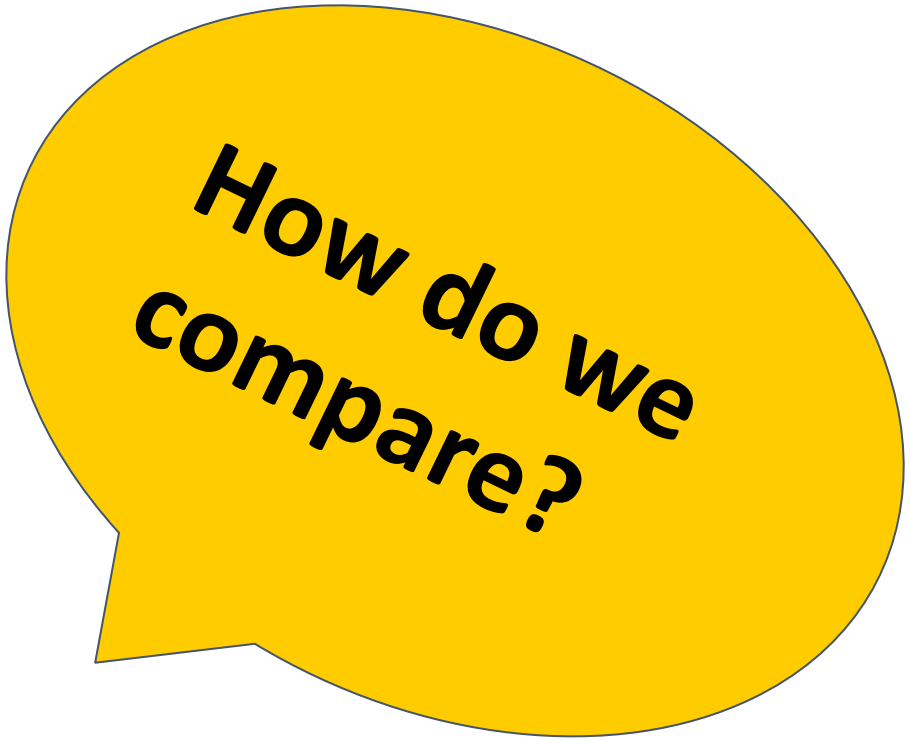
State of DevOps Report 2019



https://services.google.com/fh/files/misc/state-of-devops-2019.pdf



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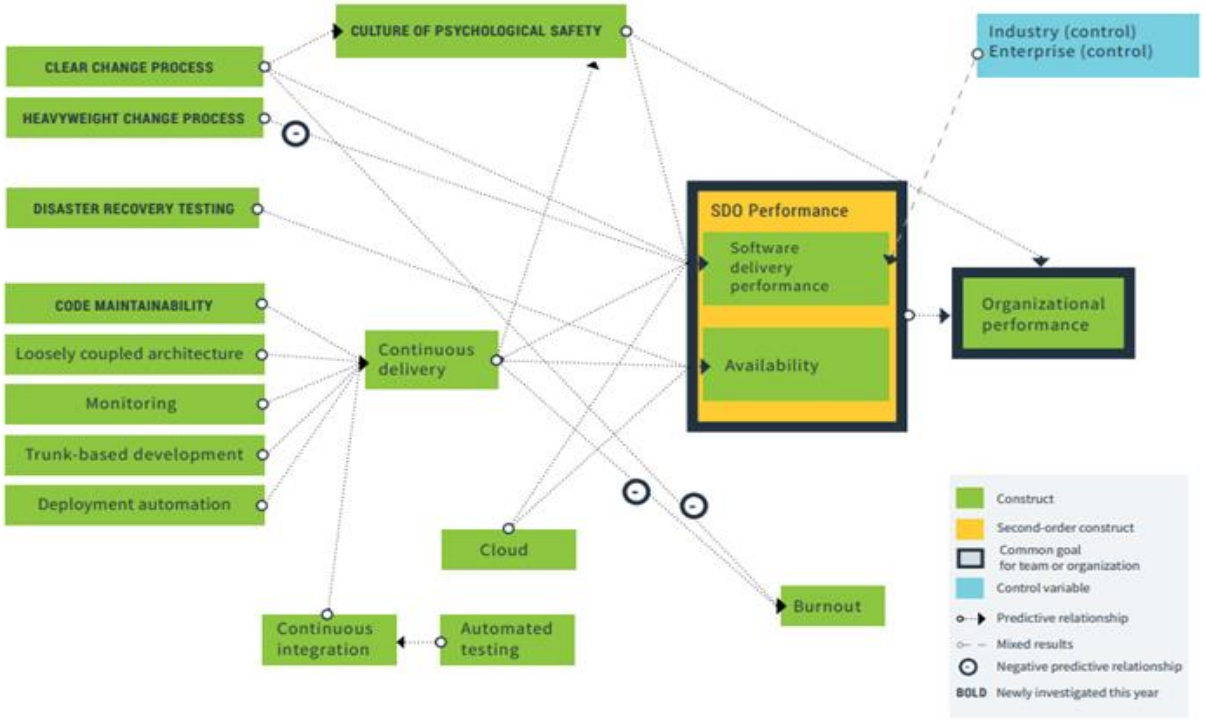
State of DevOps Report 2019



IT Performance
=>
Organization
Performance

https://services.google.com/fh/files/misc/state-of-devops-2019.pdf

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Understanding Our Challenges

Challenges

Continuous Delivery, DevOps & Data

Enterprise Data Challenges

How do we think about data?
Data lock-in

Decade old standards & conventions

Inherited data management practices

Centralize data team to design, model, deploy & operate

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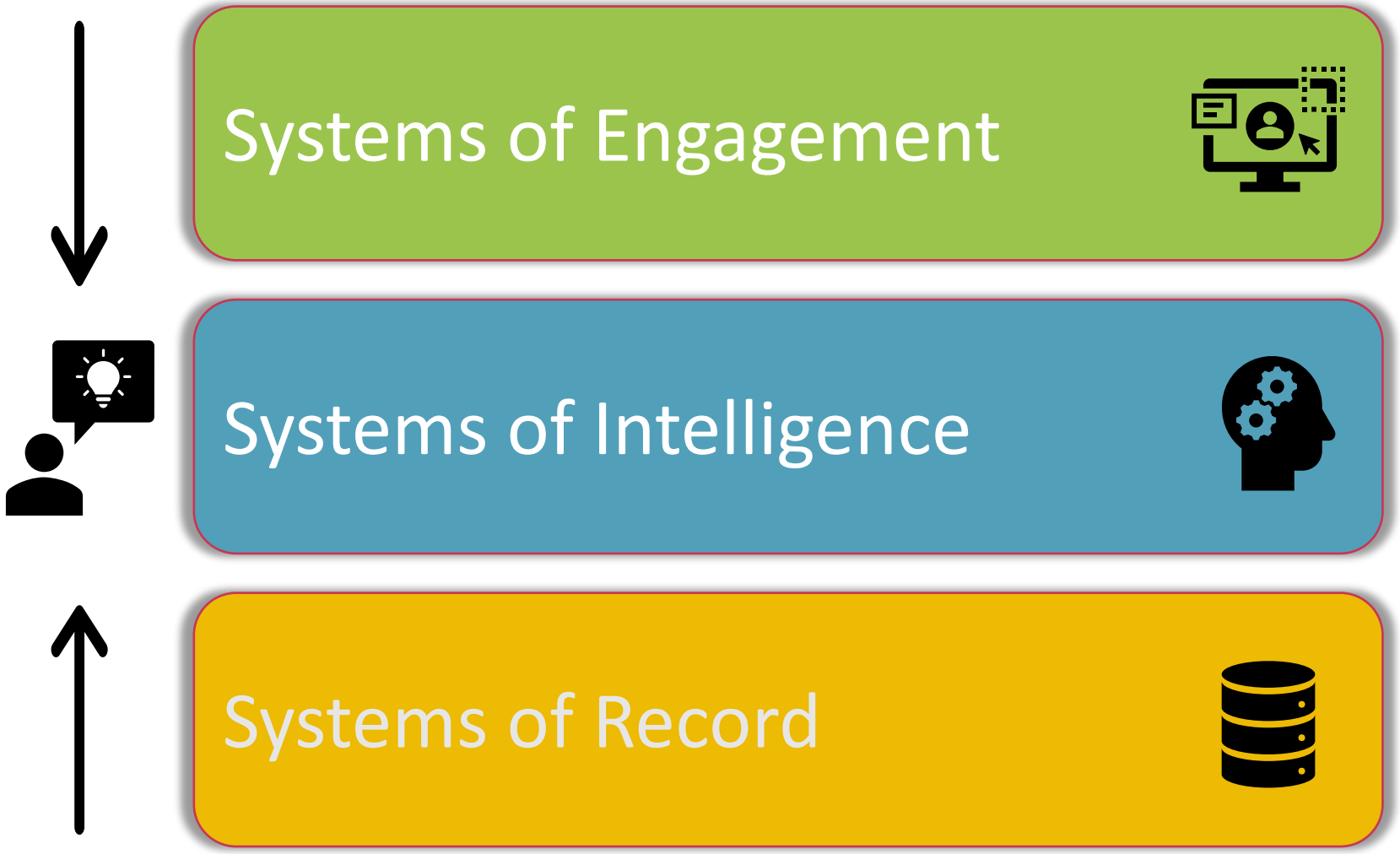
Who is Responsible?

Our system complexities are because of our thinking, thinking particularly about the "Data". We tend to compromise a lot as we could not get the "Data" in the right way, at the right time, in the proper format, to the right place


Systems of Engagement
Systems of Intelligence
Systems of Record

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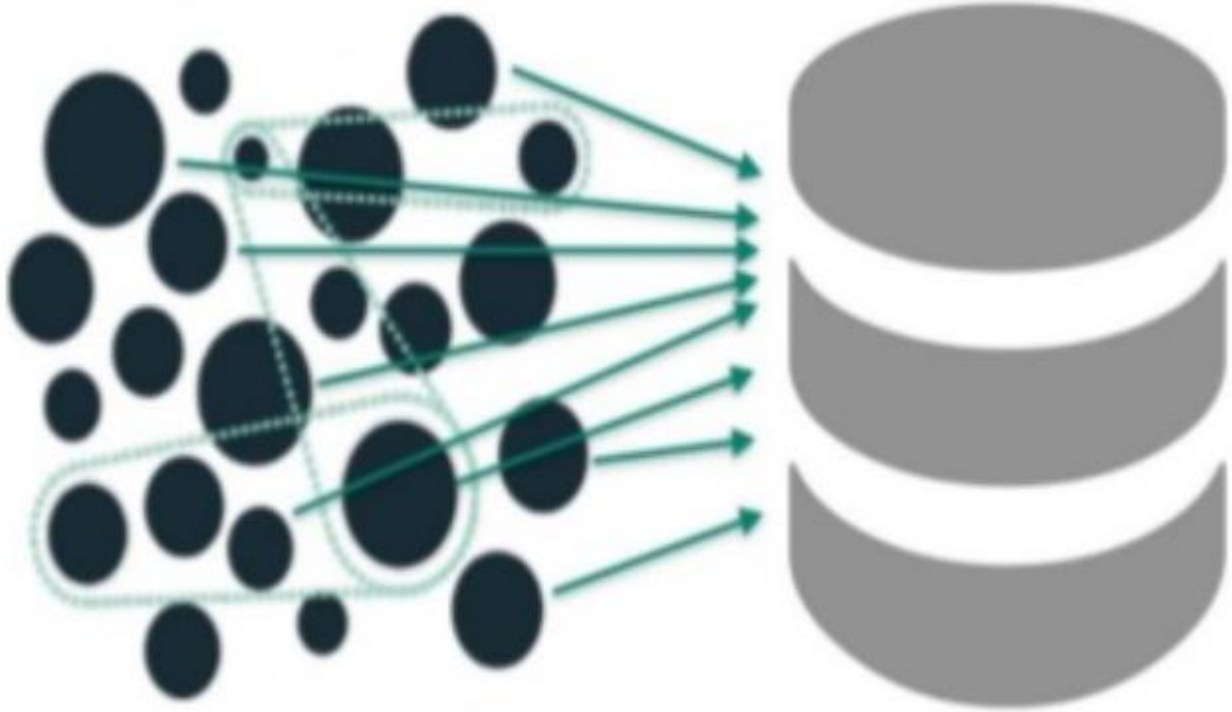
Architecture Mismatch

Microservices Era & Big Ball of "Data" mud



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Small Steps, But in Right Direction

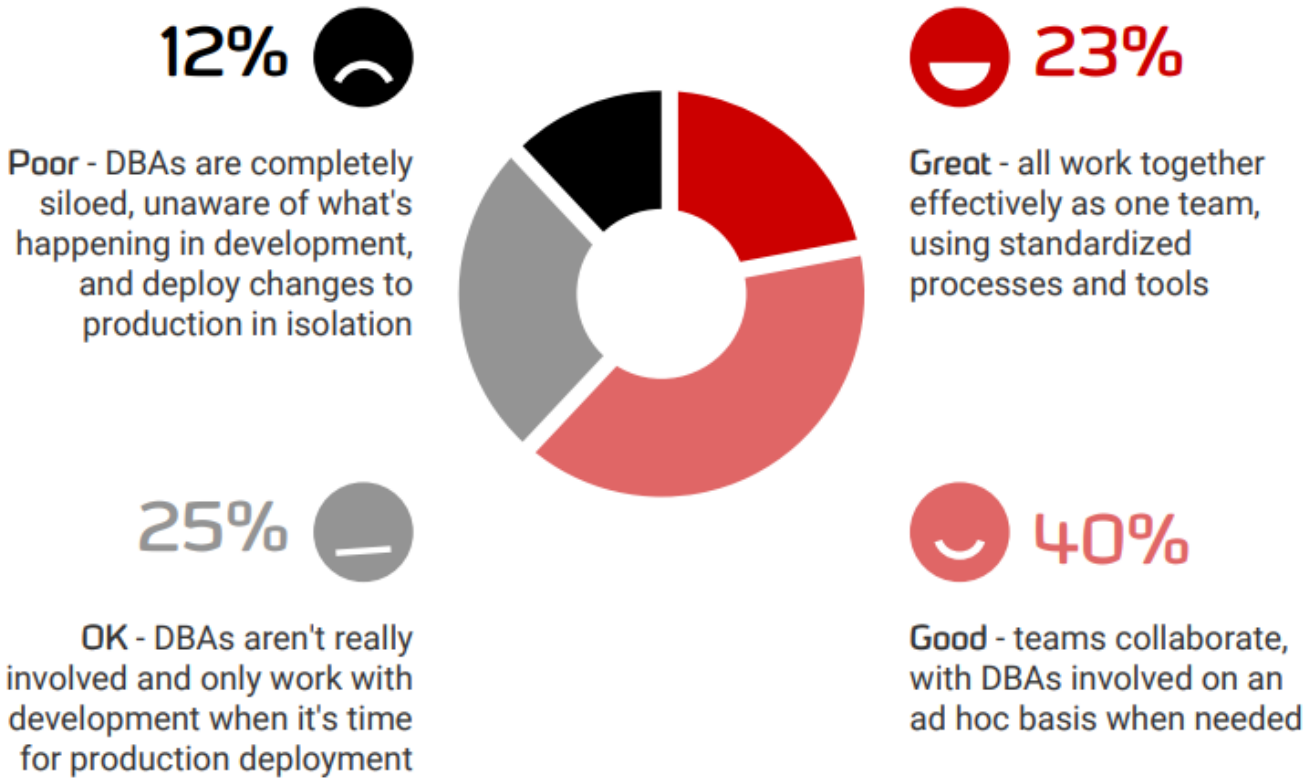
Solution Approach

Education, Conventions, Architecture, Shift Left

Solution Approach

- Decentralize Data Management Team
- Conventions & Standards
- Educate, Upskill, Tools, Processes
- Unlock the Data By Adopting Modern Practices

Solution Approach



http://assets.red-gate.com/solutions/database-devops/state-of-database-devops-2020.pdf?v=2

Help Our Data Folks - Practices

- **Make the data first-class citizen**
- **Source control - Database schema**
- **Simplify processes**
- **Review changes early**
- **CI/CD pipeline**

Help Our Data Folks - Futuristic

- **“T” & “Comb” shaped**
- **Data Engineers**
- **Educator & Mentors**
- **Platform Builders**

Help Our Dev & Ops Folks - Educate

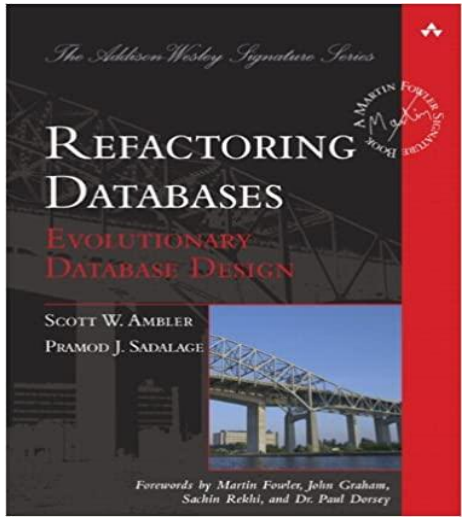
- **Data Governance, Lineage**
- **Compliance (PCI DSS, Data Privacy)**
- **Regulations (GDPR, CCPA)**
- **Data Security (PII)**
- **Standards & Conventions (Naming, model, dictionary, catalogue)**
- **Enable Self-Service (Database request, Schema change request)**

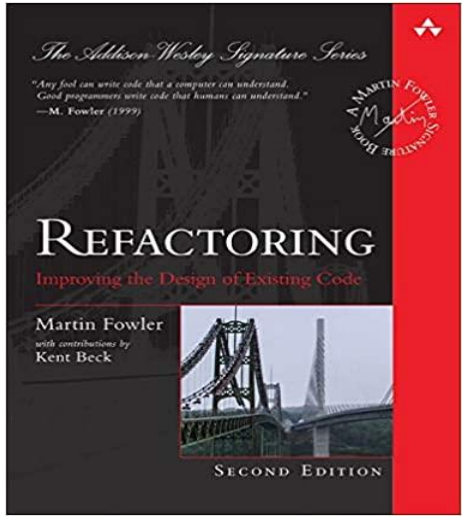
Useful Resources

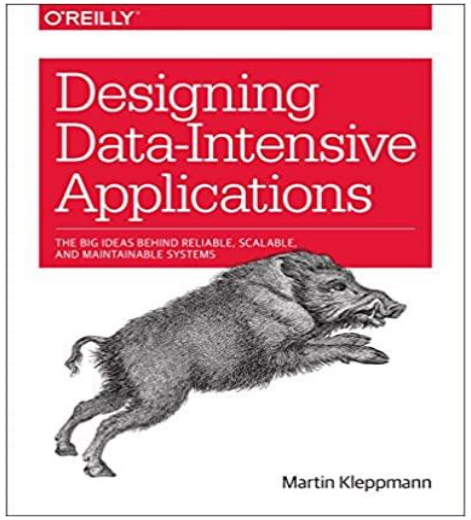
Recommendations

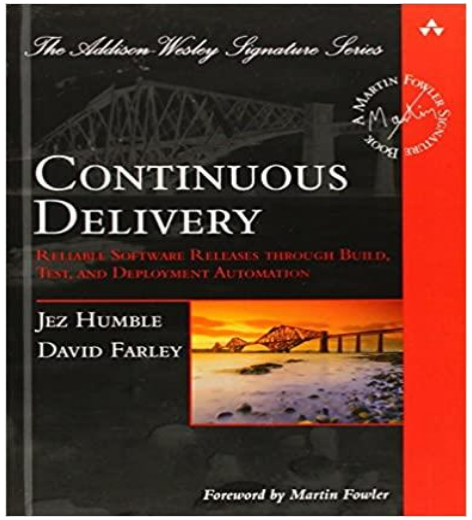
Upskill, Learn, Spread, Share

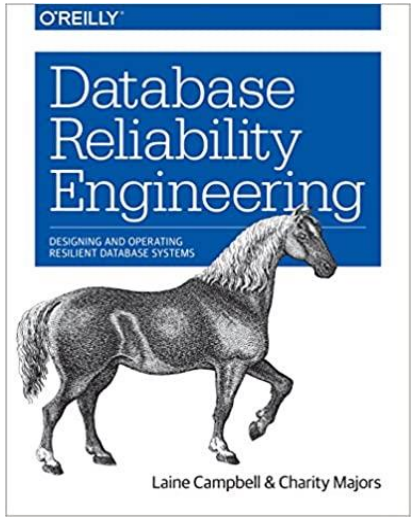
Recommendations











We need to level up our game

Data, Data Architecture, Data Management & Associated Practices need our love & attention

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Recommended Approach

- **Understanding current state (Tools, processes, roles, flow)**
- **Making work visible (current workloads, workloads in pipeline)**
- **Value Stream Management (New feature, Tech debt, Incident)**
- **Upskilling (DevOps, CI/CD Practices, Automation)**
- **Investment (Education, Time, Tools, Self-service)**
- **Role play (Data Engineer/Developer for a day)**

Collectively We Need To Come Together, Work Together, Collaborate

Remaining Challenges

Call for Action

Help – Let Us Come Together

- **Automated Governance (Including Data)**
- **Policies via automation and in pipeline**
- **Compliance as code**
- **Everyone is custodian of the Data**

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